

Revision Statement

Paper: TPAMI-2021-07-1250, "Missingness-pattern-adaptive Learning with Incomplete Data"

December 2022

Dear EIC Prof. Kyoung Mu Lee, the handling AE, and all reviewers,

We thank you for your valuable time and comments provided to improve the manuscript and our work. In this revision, we have carefully addressed all comments raised and made respective revisions in the paper. The paper has also been carefully edited and proofread. In particular, in the following,

- I: summarizes the main changes made in this revision
- II: addresses review comments by Reviewer 1
- III: addresses review comments by Reviewer 2
- IV: addresses review comments by Reviewer 3
- V: addresses review comments by Reviewer 4

Thank you very much.

Sincerely,

Yongshun Gong, Zhibin Li, Wei Liu, Xiankai Lu, Xinwang Liu, Ivor W. Tsang, Yilong Yin

I. Summary of revision

In this revision, we have made significant changes to address review comments as well as to substantially improve the paper quality:

- improved the literature review: presentation issue: more discussions are given the literature analysis, and add several recent references.
- discussed more about our technical details to address reviewers' concern.
- compared with two more state-of-the-art baselines, and added more experiments and explanations.
- more discussions are added in the parameter analysis.
- released the our source code at <https://github.com/YS-GONG/missingness-patterns>.

II. Comments made by Reviewer 1:

Recommendation: Author Should Prepare A Major Revision For A Second Review

We do appreciate your efforts and comments to help us for improving the quality of the manuscript.

Comment 1: On related work: There is no discussion between the differences of the proposed method and the methods discussed in related work. The paper could be improved by discussing the most relevant works in this section. For example, [25] is one of the methods you compare against but, surprisingly, it is mentioned only in experiments.

Response: Thank you for your advice. In our previous version, we have discussed several relevant works in the Introduction (Page 1, Lines: 49-54.), including [25] (new Ref.[27]). In this new revision, we have completed a comprehensive literature review and improved the discussions and comparisons in related work as your suggestion. Besides, four most recent studies are added and discussed in our paper. Detailed discussions are highlighted in related work (Page 3, Lines: 170-173; and Page 3, Lines: 210-234).

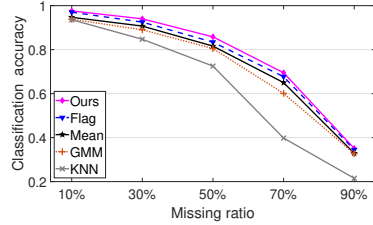
Comment 2: Experiments are not very convincing. My main critique of this work is: if you compare the performance with the other state-of-the-art algorithms (in terms of implementing them after reading this paper) in experimental section, how many researchers and practitioners will actually implement and use the proposed algorithms? This aspect would limit the impact of this work to the community. There are ways to mitigate this issue, e.g. 1) demonstrating the qualitative breakthrough of this work. And the authors should compare and analyze the performance of the proposed method and the other state-of-the-art algorithms at different incomplete rates. 2) open source the implementation.

Response: We do appreciate your comment. First, we have released our source code at <https://github.com/YS-GONG/missingness-patterns> as illustrated in abstract. Second, we have conducted experiments on datasets at different incomplete ratios, which demonstrates the superiority and generalizability of our method.

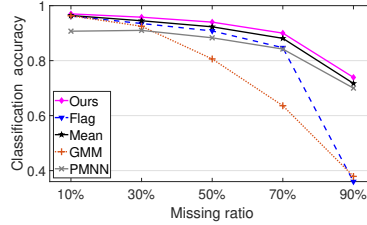
Experimental results are highlighted in [Page 11, Lines: 670-695](#). Besides, we have added two recent state-of-the-art missing data imputation methods in our linear test ([Page 8, Lines: 587-595](#)). We copy the results with varying missing ratios at below for your convenience.

We randomly removed 10%, 30%, 50%, 70%, 90% of values in them. We repeated this procedure 5 times to report the classification accuracy with mean and standard deviation.

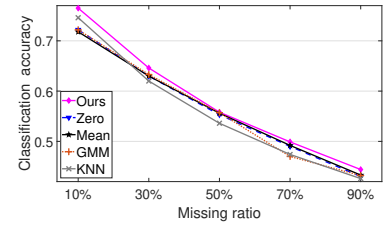
Figure 4 presents the average results on non-linear models. To keep image clear, we only draw top five methods, the comprehensive results are reported in Table 5 - 7. The tests drawn in Figure 4 demonstrate the superiority of our method with various missing ratios. Table 5 and Table 6 show the results of our method together with some baselines. The results show the advantage of our method over classical imputation methods and PMNN. Notice that PMNN produced a poor result when the missing ratio was low. PMNN is required to fit a GMM together with the neural network, but the GMM of PMNN is only trained with incomplete instances. Unlike GMM for imputation, where all data are used to fit the GMM, their model cannot be trained well when the percentage of missing is low. Flag shows good performance on Sensorless Drive Diagnosis dataset. However, its performance is limited on MNIST dataset. This indicates that the missingness patterns can be important in learning with incomplete data, but should be wisely incorporated into the model equation. Our model consistently outperforms other baselines, which verifies the effectiveness of our strategy to adjust the importance of present features by the missingness patterns.



(a) Results on dataset Sensorless Drive Diagnosis .



(b) Results on dataset MNIST.



(c) Results on dataset Avila.

Figure 4: The average accuracy.

Table 5: Classification accuracy (mean $\pm$ std) on Sensorless Drive Diagnosis dataset. The best results are bold and the second best are underlined.

Method	Percentage of missing				
	10%	30%	50%	70%	90%
Zero	0.908 $\pm$ 0.001	0.852 $\pm$ 0.011	0.769 $\pm$ 0.005	0.618 $\pm$ 0.005	0.317 $\pm$ 0.002
Mean	0.947 $\pm$ 0.004	0.907 $\pm$ 0.001	0.816 $\pm$ 0.003	0.650 $\pm$ 0.005	0.329 $\pm$ 0.003
MICE	0.717 $\pm$ 0.001	0.422 $\pm$ 0.009	0.483 $\pm$ 0.010	0.322 $\pm$ 0.007	0.197 $\pm$ 0.007
GMM	0.938 $\pm$ 0.002	0.890 $\pm$ 0.005	0.805 $\pm$ 0.007	0.601 $\pm$ 0.007	0.327 $\pm$ 0.003
KNN	0.936 $\pm$ 0.004	0.847 $\pm$ 0.003	0.725 $\pm$ 0.003	0.398 $\pm$ 0.004	0.215 $\pm$ 0.005
Flag	0.970 $\pm$ 0.001	0.925 $\pm$ 0.001	0.834 $\pm$ 0.002	0.677 $\pm$ 0.003	0.345 $\pm$ 0.004
PMNN	0.733 $\pm$ 0.001	0.886 $\pm$ 0.001	0.781 $\pm$ 0.001	0.649 $\pm$ 0.002	0.318 $\pm$ 0.001
Ours	0.976$\pm$0.001	0.940$\pm$0.001	0.858$\pm$0.002	0.695$\pm$0.002	0.351$\pm$0.002

Table 6: Classification accuracy (mean $\pm$ std) on MNIST dataset. The best results are bold and the second best are underlined.

Method	Percentage of missing				
	10%	30%	50%	70%	90%
Zero	0.957 $\pm$ 0.001	0.942 $\pm$ 0.002	0.918 $\pm$ 0.002	0.863 $\pm$ 0.003	0.688 $\pm$ 0.003
Mean	0.964 $\pm$ 0.001	<u>0.951$\pm$0.001</u>	<u>0.933$\pm$0.001</u>	<u>0.891$\pm$0.003</u>	<u>0.727$\pm$0.004</u>
GMM	0.963 $\pm$ 0.002	0.925 $\pm$ 0.003	0.806 $\pm$ 0.011	0.636 $\pm$ 0.006	0.379 $\pm$ 0.012
KNN	<u>0.965$\pm$0.001</u>	0.941 $\pm$ 0.002	0.864 $\pm$ 0.001	0.703 $\pm$ 0.023	0.223 $\pm$ 0.012
Flag	0.867 $\pm$ 0.002	0.935 $\pm$ 0.002	0.908 $\pm$ 0.003	0.847 $\pm$ 0.012	0.360 $\pm$ 0.045
PMNN	0.907 $\pm$ 0.001	0.910 $\pm$ 0.002	0.883 $\pm$ 0.003	0.842 $\pm$ 0.002	0.700 $\pm$ 0.004
Ours	0.970$\pm$0.001	0.958$\pm$0.001	0.940$\pm$0.001	0.900$\pm$0.002	0.739$\pm$0.004

Table 7: Classification accuracy (mean $\pm$ std) on Avila dataset. The best results are bold and the second best are underlined.

Method	Percentage of missing				
	10%	30%	50%	70%	90%
Zero	0.722 $\pm$ 0.002	0.630 $\pm$ 0.005	0.553 $\pm$ 0.006	<u>0.496$\pm$0.004</u>	<u>0.433$\pm$0.002</u>
Mean	0.718 $\pm$ 0.005	0.630 $\pm$ 0.005	0.556 $\pm$ 0.003	0.492 $\pm$ 0.003	0.433 $\pm$ 0.002
MICE	0.717 $\pm$ 0.005	0.618 $\pm$ 0.005	0.435 $\pm$ 0.007	0.422 $\pm$ 0.002	0.412 $\pm$ 0.001
GMM	0.722 $\pm$ 0.004	<u>0.633$\pm$0.004</u>	0.557$\pm$0.003	0.470 $\pm$ 0.002	0.432 $\pm$ 0.002
KNN	<u>0.746$\pm$0.004</u>	0.620 $\pm$ 0.002	0.536 $\pm$ 0.006	0.474 $\pm$ 0.003	0.426 $\pm$ 0.002
Flag	0.713 $\pm$ 0.005	0.630 $\pm$ 0.004	0.555 $\pm$ 0.003	0.491 $\pm$ 0.002	<u>0.433$\pm$0.002</u>
PMNN	0.503 $\pm$ 0.004	0.445 $\pm$ 0.003	0.526 $\pm$ 0.003	0.473 $\pm$ 0.004	0.412 $\pm$ 0.001
Ours	0.765$\pm$0.002	0.646$\pm$0.004	<u>0.556$\pm$0.003</u>	0.499$\pm$0.003	0.444$\pm$0.001

III. Comments made by Reviewer 2:

Recommendation: Author Should Prepare A Minor Revision

We thank the reviewer for your candid and insightful comments that have helped to improve the quality of our paper.

Comment 1: Besides minor typos or language mistakes (e.g. shortage instead of shortcomings) the paper is very well written, clear, and easy to understand.

Response: We do appreciate your positive comments and constructive advice. In this revision, we have carefully addressed all your concerns, the paper has also been carefully edited and proofread.

Comment 2: Ω^* in Equation 22 is undefined, the authors should clarify to help understanding of their proof of Lemma 5.2.

Response: Thank you for your carefully reading. Ω^* denotes the unique minimizer of $K(H)$. We have changed it to H^* for clarity (Page 7, Lines: 474-475). We copy them at below for your convenience.

$$\mathcal{U}^* = \{U : U^\top V = H^*\}, \quad (22)$$

where H^* denotes the unique minimizer of $K(H)$.

Comment 3: For clarity, the authors could justify whether the linear case $(Hm)^T x^o$ can be written in an equivalent form to the non-linear case, e.g. $\Theta^T((Hm).x^o)$ with the Hadamard product.

Response: Thank you for your great advice. The two forms could be equivalent. If we let $f(\cdot; \theta)$ in the non-linear case $\hat{y} = f((H\bar{\mathbf{m}}) \odot \mathbf{x}^o; \theta)$ be a element-wise sum function, it is then reduced to the linear case. That is, $\hat{y} = \mathbf{1}_d^\top ((H\bar{\mathbf{m}}) \odot \mathbf{x}^o)$. We have justified this in Page 5, Lines 328-331. We copy the relevant description at below for your convenience.

We can adjust the weight of observed features by missingness pattern, which gives the output:

$$\hat{y} = f((H\bar{\mathbf{m}}) \odot \mathbf{x}^o; \theta). \quad (6)$$

Considering the low-rank constraint, we decompose H by $U^\top V$. Then, the learning objective can be formulated as follows:

$$\min_{U, V, \theta} \sum_{i=1}^n \mathcal{L}(y_i, f((U^\top V \mathbf{m}_i) \odot \mathbf{x}_i^o; \theta)), \quad (7)$$

where $\mathcal{L}$ is the loss function. Note that if we let $f(\cdot; \theta)$ in the non-linear case $\hat{y} = f((H\bar{\mathbf{m}}) \odot \mathbf{x}^o; \theta)$ be a element-wise sum function, it is then reduced to the linear case. That is, $\hat{y} = \mathbf{1}_d^\top ((H\bar{\mathbf{m}}) \odot \mathbf{x}^o)$.

Comment 4: In eq. 8 it appears the function should be $\text{sign}((Hm)^T x^o)$ in order to map to -1, 1 as for SVMs.

Response: Thanks for pointing this out. You are right. We have corrected this (Page 5, Eq. 8).

$$\mathcal{F} = \{\mathbf{x} \mapsto \text{sign}((H\bar{\mathbf{m}})^T \mathbf{x}^o) : \text{rank}(H) \leq k\}. \quad (8)$$

Comments 5: The transition from problems 3 and 4 to further analyses could be made smoother by explicitly introducing the use of the hinge loss through equivalence with the $\arg \min x_{i_i}$ of the $y(m^T H^T x_i) \geq 1 - x_{i_i}, x_{i_i} \geq 0$ as for SVMs.

Response: We do appreciate your constructive advice. We have modified our equations as your suggestion (Page 4, Eq. (3) and Page 5, Eq. (4)). We copy the new problem expressions at below for your convenience.

$$\min_H \quad \frac{1}{n} \|M \odot (H\bar{M})\|_F^2 + \eta_1 \|H\|_F^2 + \frac{\eta_2}{n} \sum_{i=1}^n \ell(y_i, \bar{\mathbf{m}}_i^T H^T \mathbf{x}_i^o), \quad (3)$$

$$\text{s.t.} \quad \text{rank}(H) \leq k, M = [\mathbf{m}_1, \dots, \mathbf{m}_n], \bar{M} = [\bar{\mathbf{m}}_1, \dots, \bar{\mathbf{m}}_n],$$

where $\ell(y, \hat{y}) \triangleq \max(0, 1 - y\hat{y})$ denotes the hinge loss.

$$\min_{U, V} \quad \frac{1}{n} \|M \odot (U^T V \bar{M})\|_F^2 + \eta_1 \|U^T V\|_F^2 + \frac{\eta_2}{n} \sum_{i=1}^n \ell(y_i, \bar{\mathbf{m}}_i^T V^T U \mathbf{x}_i^o), \quad (4)$$

$$\text{s.t.} \quad M = [\mathbf{m}_1, \dots, \mathbf{m}_n], \bar{M} = [\bar{\mathbf{m}}_1, \dots, \bar{\mathbf{m}}_n].$$

Comments 6: Although this absence seems justified in light of [44] (no spurious local minima for low-rank matrix factorization), how did the authors initialize their algorithm and did they observe any impact on practical implementations?

Response: We do appreciate your question. As we see results in Table 5-7 in the paper, the standard deviations are very small with various missing ratios. The model parameters in our experiments were all randomly initialised and the impact is little as we could see.

Comments 7: On the topic of implementation and optimization algorithm: given the similarity of their algorithm for linear classifiers to SVMs, could the authors justify their choice of a subgradient method, compared to primal-dual formulations or coordinate descent on the dual, as used for SVMs?

Response: Thank you for your question. Because the proposed algorithm w.r.t its parameters is not convex, primal-dual formulations or coordinate descent on the dual is not feasible, at least not in an easy way. That is why we choose to use subgradient method.

Table 8: Linear model vs. non-linear model.

Models	Datasets				
	bands	hepatitis	horse	mammographic	pima
linear	0.681±0.004	0.872±0.000	0.879±0.006	0.799±0.001	0.791± 0.002
non-linear	0.689±0.005	0.874±0.000	0.886±0.003	0.812±0.002	0.797± 0.001

Comments 8: The authors justify through the form of the error bound and grid search on the rank constraint value k that low-rankness improves performance. However, they do not provide a comparison with the full-rank convex problem, and k does not go up to 19 and 22 in Fig. 5.

Response: We do appreciate your comment. We have added new plots in Fig. 5 and 6 (Page 11) for a wider range of k , including results on full-rank convex problem.

Comments 9: The authors further claimed their method is insensitive to k , this seems contradictory with claiming the low-rank structure is beneficial (see e.g. 1691, "insensitive to k but the smaller the better" is a contradiction). In Fig 5. for the bands dataset, $k = 18$ and $k = 2$ seem to perform very similarly.

Response: Thanks for pointing this out. We said "insensitive to k but the smaller the better", because the k is an upper bound of the rank, which means the rank of the learned matrix can be smaller than k . For example, if the optimal rank is 2, even we set k to some values greater than 2, the learned matrix rank could still be approximately 2 until training data is not enough to recover this optimal rank. Thus, the results were stable up to some k 's. Low-rankness in practice can be caused by 1) some of the original features are redundant or have little effect on the task and/or 2) some of the features are highly correlated w.r.t the task. As a result, columns in the optimal H should correlated. The learned H then becomes rank-deficient naturally. Anyway, smaller k will be better if we consider it helps reduce the number of parameters by the factorization $H = U^\top V$. We have also added the plots on accuracy w.r.t k in Fig.5 and 6 to make this clearer with corresponding discussions.

Comments 10: Given these observations, and that the full-rank problem involving only H can be expected to be easier to optimize in practice, an experimental comparison for the linear and non-linear cases would have been interesting.

Response: We do appreciate your advice. We have added an experiment for the linear and non-linear model comparison. Our proposed non-linear model is also under the low-rank constraint as shown in Eq. 7. As the results shown in Table 8 in this revision statement, non-linear model achieves a slightly better result compared to the linear one.

Table 9: Missing completely at random vs. Missing with rules.

Models	Datasets				
	bands	hepatitis	horse	mammographic	pima
Missing at random	0.648 $\pm$ 0.021	0.868 $\pm$ 0.009	0.840 $\pm$ 0.011	0.776 $\pm$ 0.010	0.756 $\pm$ 0.006
Missing with patterns	0.664 $\pm$ 0.016	0.872 $\pm$ 0.010	0.844 $\pm$ 0.009	0.792 $\pm$ 0.005	0.758 $\pm$ 0.006

Comments 11: The authors claim the method is adaptive to missingness patterns in the data. Could the authors clarify how the additional missing values are introduced in their experiments? Do they follow a pattern? I would like to see experiments using structured sparsity or other pattern-following introduction of missing values in the data.

Response: Thank you for your question and advice. In our experiments, we removed values completely at random, which indicates that the data are missing is independent of the observed and unobserved data. To address your concern and verify the structured missing patterns, we provide an experiment that systematically removes values under designed rules. For example, there are three missing rules in the bands dataset: (1) if the second feature is missing, the fourth feature has 50% chance to miss; (2) if the 8th feature is missing, the 11th feature has 50% chance to miss; (3) if the 14th feature is missing, the 16th feature has 50% chance to miss. The random missing ratio and the structured missing ratio are around 30%.

As we can see from Table 9, if the data are missing is systematically related to the unobserved data, that is, there are missingness patterns in the data, our model can capture such patterns and perform better than the data missing completely at random. We have added this experiment in the Appendix.

Comments 12: In the presence of missingness patterns vs fully random missing values, how does the learn dictionary differ? Can analysis of the dictionary reveal missigness patterns a posteriori?

Response: We do appreciate your advice. We designed two new experiments to address your question.

- **First experiment and steps:** (1) we train our model 500 times on datasets that missing completely at random (30%), then we can generate 500 dictionaries as positive samples that learned from such data; (2) we first generate datasets that missing completely at random (30%), and then add missing rules (e.g., if the second feature is missing, the fourth feature has 50% chance to miss) to these datasets. Our model are trained 500 times on the above datasets that 500 dictionaries can be generated as negative samples; (3) we use a simple SVM classifier to test whether learned dictionaries contain posteriori knowledge that can make a distinction between missing at random and missing with rules.

Table 10: Dictionary Classification Accuracy (mean $\pm$ std) on datasets.

	Dictionaries Generated From:				
	bands	hepatitis	horse	mammographic	pima
First experiment	0.938 $\pm$ 0.012	0.948 $\pm$ 0.016	0.914 $\pm$ 0.048	0.960 $\pm$ 0.010	0.941 $\pm$ 0.013
Second experiment	0.957 $\pm$ 0.009	0.942 $\pm$ 0.010	0.951 $\pm$ 0.011	0.986 $\pm$ 0.004	0.970 $\pm$ 0.006

• **Second experiment and steps:** (1) we first generate datasets that missing completely at random (30%); (2) and then we add different missing rules to these datasets. There are two different datasets under different missing rules; (3) we train the model 500 times on these two datasets respectively, that we can get 500 dictionaries as positive samples and 500 dictionaries as negative ones. (4) We use a SVM classifier to test whether the learned dictionaries can be distinguished when they generated from different missingness rules.

As the results reported in Table 10, the learned dictionaries are distinguishable in our designed two experiments. The results in the first experiment indicate that the learned dictionary contain missingness knowledge. Dictionaries are different when they following the random missing and specific missing patterns. The results in the second experiment demonstrate that the learned dictionaries can also be distinguished when they generated from different missingness rules.

We have added the above experiments in the Appendix.

Comments 13: Finally, for neural networks, the authors should clarify how many additional parameters come from using their method and how significant this number is compared to the base models.

Response: Thank you for your great advice. Compared to base models, our non-linear model only introduces two learnable matrices $U \in \mathbb{R}^{k \times d}$ and $V \in \mathbb{R}^{k \times 2d}$, where d is the numbers of dimensions of the sample $\mathbf{x}$. The number of extra parameters occupied a small proportion of the entire neural network. We have added the above discussions in Page 5, Lines 338-341.

IV. Comments made by Reviewer 3:

Recommendation: Author Should Prepare A Minor Revision

We thank the reviewer for your time and efforts in providing a detailed assessment of our paper.

Comment 1: It seems it is not required to consider $\bar{m}$. same result will be produced using M in place of $\bar{M}$ and keeping H is a matrix $d \times d$.

Response: Thank you for your question. The $\bar{m}$ would help include some compensation for missing features by introducing corresponding vectors in H , so that no matter how many features are missing, there

is always the same number of components chosen from H . This reduces the bias of using only $\mathbf{m}$. This is similar to onehot encoding of many 2-category features (missing/observed). If there is no missing feature, the expansion of $\mathbf{m}$ will be redundant. If there are missing features, this will also affect the coefficients for observed features. To see this, assume $\mathbf{w} = H\bar{\mathbf{m}}$, if the features are all observed, then $\mathbf{w} = \mathbf{h}_1 + \mathbf{h}_2 + \mathbf{h}_3$. If x_2 is missing, then $\mathbf{w} = \mathbf{h}_1 + \mathbf{h}_{2+d} + \mathbf{h}_3$. Missing of x_2 will replace the component $\mathbf{h}_2$ with $\mathbf{h}_{2+d}$, so it has effect to every bit of $\mathbf{w}$, including the coefficients for observed features.

V. Comments made by Reviewer 4:

Recommendation: Author Should Prepare A Major Revision For A Second Review

We do appreciate your efforts and comments to help us for improving the quality of the manuscript.

Comment 1: The motivation to expand the “ $\mathbf{m}$ ” to “ $\hat{\mathbf{m}}$ ” is not clear. I can roughly get the idea of avoiding zero results for the missing patterns. However, will this operation affect the pattern that is not missing?

Response: Thank you once again for your review and question. The $\bar{\mathbf{m}}$ would help include some compensation for missing features by introducing corresponding vectors in H , so that no matter how many features are missing, there is always the same number of components chosen from H . This reduces the bias of using only $\mathbf{m}$. This is similar to onehot encoding of many 2-category features (missing/observed). If there is no missing feature, the expansion of $\mathbf{m}$ will be redundant. If there are missing features, this will also affect the coefficients for observed features. To see this, assume $\mathbf{w} = H\bar{\mathbf{m}}$, if the features are all observed, then $\mathbf{w} = \mathbf{h}_1 + \mathbf{h}_2 + \mathbf{h}_3$. If x_2 is missing, then $\mathbf{w} = \mathbf{h}_1 + \mathbf{h}_{2+d} + \mathbf{h}_3$. Missing of x_2 will replace the component $\mathbf{h}_2$ with $\mathbf{h}_{2+d}$, so it has effect to every bit of $\mathbf{w}$, including the coefficients for observed features.

Comment 2: The objective is given In Eq. (3), which is to minimize two Frobenius norm items, but what are the purposes?

Response: We do appreciate your question. Similar to that in SVM, the first term in Eq. (3) is the approximate denominator for instance-based margins. We borrow this idea from [16] to ensure a fair optimization of margins w.r.t each instance. Similarly, the second term is aimed for better generalisation across training/test and stable optimisation. This can be seen from proof of Lemma 5.2 that it guarantees the strongly convexity of the loss function.

Comment 3: In the experimental part, the baselines are composed of some early methods, like one published in 2008. I am more curious about the latest studies in this field.

Response: Thank you for your advice. In this revision, we have added two latest studies in our experiments. One method was designed to impute gene expression matrices via a biased low-rank matrix decomposition approach [1] (2021, *Briefings in Bioinformatics*). The other method is a state-of-the-art missing data imputation method based on the spatio-temporal correlation learning [2] (2021, *TKDE*). The method description are shown in Page 8, Linse 587-595. The comparison results are reported in Table 3 and 4. We copy the Tables and discussions at below for your convenience.

We present experiments with the original datasets here in Table 3. The original datasets contains internally absent attributes. Our model consistently outperforms other baselines except on pima dataset. The performance gaps between all models are relatively small due to the low missing percentages.

Experiment results in the general case (with additionally 30% data removed) are presented in Table 4. We repeated the experiments 5 times to report the classification accuracy. Our method achieved the best accuracy on all 5 datasets. In general, our method is better than imputation methods, because inaccurate imputation could deteriorate the downstream classification task. Our method also outperforms Flag, which indicates that simply adding the missingness pattern as additional features is not as good as our strategy. These datasets contain inherent missing features, and we also removed some values randomly. These factors make the missingness mechanism complicated and it is hard to learn a universal model that fits all these heterogeneous missingness patterns. Our method tries to adaptively apply the classifiers specialized to different missingness patterns, which makes it capable of learning some non-linear classifiers. This makes our model outperforms other baselines.

Table 3: Classification accuracy (mean $\pm$ std) with original datasets. The best results are bold and the second best are underlined.

Method	Dataset					
	bands	hepatitis	horse	mammographic	pima	MIC
Flag	0.617 $\pm$ 0.000	0.872$\pm$0.000	0.864 $\pm$ 0.000	0.778 $\pm$ 0.000	0.783 $\pm$ 0.000	0.771 $\pm$ 0.003
Zero	0.606 $\pm$ 0.002	0.851 $\pm$ 0.000	0.838 $\pm$ 0.000	0.796 $\pm$ 0.000	<u>0.801$\pm$0.000</u>	0.763 $\pm$ 0.005
Mean	0.611 $\pm$ 0.002	0.872$\pm$0.000	0.847 $\pm$ 0.000	0.796 $\pm$ 0.000	0.792 $\pm$ 0.000	0.765 $\pm$ 0.004
MICE	0.617 $\pm$ 0.000	0.809 $\pm$ 0.000	0.856 $\pm$ 0.000	0.785 $\pm$ 0.000	0.775 $\pm$ 0.000	0.763 $\pm$ 0.001
GMM	0.594 $\pm$ 0.013	0.872$\pm$0.021	0.841 $\pm$ 0.011	0.779 $\pm$ 0.010	0.787 $\pm$ 0.023	0.752 $\pm$ 0.003
KNN	0.593 $\pm$ 0.010	0.847 $\pm$ 0.009	0.847 $\pm$ 0.000	0.775 $\pm$ 0.002	0.805$\pm$0.000	<u>0.773$\pm$0.003</u>
geom	0.605 $\pm$ 0.000	0.872$\pm$0.000	<u>0.865$\pm$0.000</u>	0.789 $\pm$ 0.000	0.792 $\pm$ 0.000	0.758 $\pm$ 0.005
karma	0.611 $\pm$ 0.040	0.809 $\pm$ 0.000	0.838 $\pm$ 0.000	<u>0.798$\pm$0.000</u>	0.797 $\pm$ 0.005	0.767 $\pm$ 0.002
WEDGE	<u>0.609$\pm$0.015</u>	<u>0.812$\pm$0.004</u>	<u>0.841$\pm$0.010</u>	<u>0.790$\pm$0.002</u>	<u>0.783$\pm$0.002</u>	<u>0.762$\pm$0.001</u>
SNM-NMF	<u>0.624$\pm$0.003</u>	<u>0.850$\pm$0.000</u>	<u>0.865$\pm$0.000</u>	0.778 $\pm$ 0.000	0.797 $\pm$ 0.008	0.765 $\pm$ 0.002
Ours	0.678$\pm$0.005	0.872$\pm$0.000	0.876$\pm$0.007	0.799$\pm$0.001	0.791 $\pm$ 0.002	0.778$\pm$ 0.000

Comment 4: The operation of “zero out” should be clarified. I guess it is operated like padding zero into those missing positions.

Table 4: Classification accuracy (mean $\pm$ std) on datasets with additional missing values. The best results are bold and the second best are underlined.

Method	Dataset					
	bands	hepatitis	horse	mammographic	pima	MIC
Flag	0.583 $\pm$ 0.006	<u>0.845$\pm$0.016</u>	0.825 $\pm$ 0.007	0.764 $\pm$ 0.006	0.737 $\pm$ 0.022	<u>0.773$\pm$0.004</u>
Zero	0.597 $\pm$ 0.022	0.842 $\pm$ 0.017	0.816 $\pm$ 0.015	0.761 $\pm$ 0.018	0.736 $\pm$ 0.010	0.759 $\pm$ 0.003
Mean	0.586 $\pm$ 0.008	0.843 $\pm$ 0.017	0.822 $\pm$ 0.013	<u>0.774$\pm$0.009</u>	0.740 $\pm$ 0.002	0.761 $\pm$ 0.003
MICE	0.575 $\pm$ 0.027	0.774 $\pm$ 0.044	0.712 $\pm$ 0.041	<u>0.772$\pm$0.016</u>	0.738 $\pm$ 0.023	0.764 $\pm$ 0.002
GMM	0.572 $\pm$ 0.021	0.825 $\pm$ 0.037	0.805 $\pm$ 0.013	0.768 $\pm$ 0.012	0.742 $\pm$ 0.021	0.764 $\pm$ 0.005
KNN	0.592 $\pm$ 0.016	0.812 $\pm$ 0.037	<u>0.836$\pm$0.026</u>	0.762 $\pm$ 0.006	<u>0.747$\pm$0.009</u>	0.771 $\pm$ 0.005
geom	0.575 $\pm$ 0.023	0.834 $\pm$ 0.025	0.819 $\pm$ 0.022	0.762 $\pm$ 0.009	<u>0.742$\pm$0.006</u>	0.764 $\pm$ 0.002
karma	0.551 $\pm$ 0.040	0.817 $\pm$ 0.032	0.759 $\pm$ 0.022	0.759 $\pm$ 0.014	0.740 $\pm$ 0.009	0.751 $\pm$ 0.008
WEDGE	<u>0.566$\pm$0.017</u>	<u>0.844$\pm$0.021</u>	<u>0.819$\pm$0.031</u>	<u>0.763$\pm$0.012</u>	<u>0.738$\pm$0.017</u>	<u>0.770$\pm$0.003</u>
SVM-NMF	<u>0.603$\pm$0.010</u>	<u>0.838$\pm$0.024</u>	<u>0.827$\pm$0.014</u>	<u>0.774$\pm$0.010</u>	<u>0.743$\pm$0.008</u>	<u>0.770$\pm$0.006</u>
Ours	0.648$\pm$0.021	0.868$\pm$0.009	0.840$\pm$0.011	0.776$\pm$0.010	0.756$\pm$0.006	0.780$\pm$0.004

Response: Thank you for pointing out this problem. Zero-out means filling missing values with zeros. We have clarified this in the new revision (Page 3, Lines: 254-255 and Page 4, Lines: 293-295). We copy them at below for your convenience.

where $\mathbf{x}^o \in \mathbb{R}^d$ denotes the $\mathbf{x}$ after zero-filling for the missing values.

This is achieved through a mask matrix M to set the weights to zeros in $H\bar{M}$ corresponding to missing features.

Comment 5: In Eq. (3), the learning parameters are not only “H” but also “ η ”.

Response: Thank you for your question. As the description of Eq. (3), η_1 , η_2 and k are hyper-parameters (Page 4, Lines: 282-283); In our method, we fixed $\eta_1 = 10^{-6}$ for our method. η_2 and k were chosen from $\{10^{-5}, \dots, 10^5\}$ and $\{2, 4, \dots, d\}$ where d is the feature dimension of related dataset (Page 9, Lines: 601-604). Besides, the analysis of parameters are shown in section 6.3.

References

- [1] Y. Hu, B. Li, W. Zhang, N. Liu, P. Cai, F. Chen, and K. Qu, “Wedge: imputation of gene expression values from single-cell rna-seq datasets using biased matrix decomposition,” *Briefings in Bioinformatics*, vol. 22, no. 5, p. bbab085, 2021.
- [2] Y. Gong, Z. Li, J. Zhang, W. Liu, Y. Yin, and Y. Zheng, “Missing value imputation for multi-view urban statistical data via spatial correlation learning,” *IEEE Transactions on Knowledge and Data Engineering*, 2021.